

A Concise Derivation of Laplacian Eigenmaps

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Goal

Given n high-dimensional data points

$$x_i \in \mathbb{R}^p, \quad i = 1, \dots, n,$$

Laplacian Eigenmaps (LE) seeks a *low-dimensional* embedding

$$y_i \in \mathbb{R}^k, \quad k \ll p,$$

that **preserves local geometric structure**: points that are close in the original space should remain close in the embedding.

The core idea is to encode *local proximity* via a weighted graph, then find an embedding that minimises a weighted sum of squared pairwise distances—pulling nearby points together.

Step 1: Construct the Neighbourhood Graph

Why? We want to capture only *local* structure; global distances in high dimensions are unreliable (curse of dimensionality).

Connect each x_i to its K nearest neighbours (or to all points within a ball of radius ε) to form an undirected graph $G = (V, E)$ with vertex set $V = \{1, \dots, n\}$ and edge set E .

Assign edge weights using the **heat kernel**:

$$w_{ij} = \begin{cases} \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) & \text{if } (i, j) \in E, \\ 0 & \text{otherwise,} \end{cases}$$

where $\sigma > 0$ is a bandwidth parameter. A larger w_{ij} indicates that x_i and x_j are more similar; a smaller distance yields a weight closer to 1.

Collect the weights into the **weight matrix** $W \in \mathbb{R}^{n \times n}$, where $W_{ij} = w_{ij}$.

Step 2: Formulate the Objective

Why? To preserve locality, we penalise configurations where nearby points (large w_{ij}) are mapped far apart.

The objective to *minimise* is

$$\mathcal{L}(Y) = \frac{1}{2} \sum_{i,j} w_{ij} \|y_i - y_j\|^2,$$

where $Y = [y_1 \mid \dots \mid y_n]^T \in \mathbb{R}^{n \times k}$ collects all embedding vectors as rows.

A large w_{ij} forces the corresponding term $\|y_i - y_j\|^2$ to be small, exactly encoding the “nearby points stay nearby” requirement.

Step 3: Simplify with the Graph Laplacian

Why? Expanding the quadratic form reveals a clean matrix expression involving two standard graph matrices.

Expanding the objective (one output dimension ℓ): Using $\|y_i - y_j\|^2 = \sum_{\ell=1}^k (Y_{i\ell} - Y_{j\ell})^2$,

$$\mathcal{L}(Y) = \frac{1}{2} \sum_{\ell=1}^k \sum_{i,j} w_{ij} (Y_{i\ell} - Y_{j\ell})^2.$$

Focus on a single column $f = Y_{:\ell} \in \mathbb{R}^n$. Expanding,

$$\begin{aligned} \frac{1}{2} \sum_{i,j} w_{ij} (f_i - f_j)^2 &= \frac{1}{2} \sum_{i,j} w_{ij} (f_i^2 - 2f_i f_j + f_j^2) \\ &= \sum_{i,j} w_{ij} f_i^2 - \sum_{i,j} w_{ij} f_i f_j \\ &= \sum_i d_i f_i^2 - f^T W f, \end{aligned}$$

where we define the **degree** of node i as

$$d_i = \sum_j w_{ij},$$

and the **degree matrix** $D \in \mathbb{R}^{n \times n}$ as the diagonal matrix with $D_{ii} = d_i$.

The expression $\sum_i d_i f_i^2 = f^T D f$, so

$$\frac{1}{2} \sum_{i,j} w_{ij} (f_i - f_j)^2 = f^T D f - f^T W f = f^T (D - W) f.$$

Define the **(unnormalised) graph Laplacian**

$$L = D - W \in \mathbb{R}^{n \times n}.$$

Summing over all k output dimensions,

$$\mathcal{L}(Y) = \sum_{\ell=1}^k Y_{:\ell}^T L Y_{:\ell} = \text{tr}(Y^T L Y).$$

The objective therefore reduces to

$$\min_{Y \in \mathbb{R}^{n \times k}} \text{tr}(Y^T L Y).$$

Step 4: Add Constraints to Prevent Degenerate Solutions

Why? Without constraints, $Y = 0$ trivially minimises the objective; moreover, the embedding is defined only up to rotation and translation (non-uniqueness). Two constraints are imposed.

Constraint 1 (scale). Fix the spread of the embedding to avoid collapse to a point:

$$Y^T D Y = I_k,$$

where I_k is the $k \times k$ identity matrix. The degree matrix D acts as a measure: nodes with higher total weight (more or stronger connections) carry more “mass”.

Constraint 2 (translation / constant removal). The all-ones vector $\mathbf{1}$ lies in the null space of L (since $L\mathbf{1} = 0$), corresponding to the trivial constant embedding. We exclude it by requiring each embedding coordinate to be D -orthogonal to $\mathbf{1}$:

$$Y^T D \mathbf{1} = 0.$$

The constrained optimisation problem is therefore

$$\min_{Y \in \mathbb{R}^{n \times k}} \text{tr}(Y^T L Y) \quad \text{subject to} \quad Y^T D Y = I_k, \quad Y^T D \mathbf{1} = 0.$$

Step 5: Reduce to a Generalised Eigenvalue Problem

Why? The constrained problem is a *trace-minimisation under an orthogonality constraint*, a classical problem whose solution is given by the generalised eigenvectors with the smallest eigenvalues.

Applying the Lagrangian method. For a single embedding direction $f \in \mathbb{R}^n$ (one column of Y), the problem is

$$\min_f f^T L f \quad \text{subject to} \quad f^T D f = 1.$$

Utilising the method of Lagrange multipliers (see Appendix A), form the Lagrangian

$$\mathcal{F}(f, \lambda) = f^T L f - \lambda(f^T D f - 1).$$

Taking the derivative with respect to f and setting it to zero (see Appendix B):

$$\frac{\partial \mathcal{F}}{\partial f} = 2L f - 2\lambda D f = 0 \implies L f = \lambda D f.$$

This is a **generalised eigenvalue problem**:

$$L f = \lambda D f.$$

Why the smallest eigenvalues? At a solution $L f = \lambda D f$, the objective value satisfies

$$f^T L f = \lambda \underbrace{f^T D f}_{=1} = \lambda.$$

Minimising $f^T L f$ is therefore equivalent to finding the *smallest* eigenvalues λ .

Excluding the trivial solution. L is positive semi-definite (see Appendix C) with $\mathbf{1}$ as a zero eigenvector; $\lambda_0 = 0$ gives the trivial constant embedding. The Constraint 2 from Step 4 ($Y^T D \mathbf{1} = 0$) removes this solution.

Solution. Solve $Lf = \lambda Df$ and sort the eigenvalues

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{n-1}.$$

Let $f_1, f_2, \dots, f_k$ be the eigenvectors corresponding to $\lambda_1, \dots, \lambda_k$ (the k smallest *non-zero* eigenvalues), each normalised so that $f_j^T Df_j = 1$.

Conclusion

The Laplacian Eigenmaps embedding is

$$y_i = \begin{bmatrix} f_1(i) \\ f_2(i) \\ \vdots \\ f_k(i) \end{bmatrix} \in \mathbb{R}^k, \quad i = 1, \dots, n,$$

where $f_j(i)$ denotes the i -th component of eigenvector f_j .

The derivation proceeds as follows:

1. Build a weighted neighbourhood graph to encode local proximity via W and D .
2. Minimise $\text{tr}(Y^T LY)$, the weighted sum of squared pairwise distances in the embedding, where $L = D - W$.
3. Constrain $Y^T DY = I_k$ (scale) and $Y^T D\mathbf{1} = 0$ (remove trivial solution).
4. Solve the generalised eigenvalue problem $Lf = \lambda Df$; the k eigenvectors with the smallest non-zero eigenvalues form the embedding.

Laplacian Eigenmaps preserves local structure by embedding data into the eigenvectors of the graph Laplacian $L = D - W$ corresponding to the k smallest non-zero eigenvalues of $Lf = \lambda Df$.

Appendix A: Method of Lagrange Multipliers

Theorem. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable. At a constrained extremum of $f(x)$ subject to $g(x) = 0$, there exists a scalar $\lambda \in \mathbb{R}$ (the *Lagrange multiplier*) such that

$$\nabla f(x) = \lambda \nabla g(x).$$

Application here. With $f(v) = v^T Lv$ and $g(v) = v^T Dv - 1 = 0$, the stationarity condition gives

$$\nabla_v(v^T Lv) = \lambda \nabla_v(v^T Dv) \implies 2Lv = \lambda \cdot 2Dv \implies Lv = \lambda Dv.$$

Appendix B: Derivative of a Quadratic Form

For a symmetric matrix $M \in \mathbb{R}^{n \times n}$ and $v \in \mathbb{R}^n$,

$$\frac{\partial}{\partial v}(v^T Mv) = 2Mv.$$

Derivation.

$$v^T M v = \sum_{i,j} M_{ij} v_i v_j.$$

$$\frac{\partial}{\partial v_k} (v^T M v) = \sum_j M_{kj} v_j + \sum_i M_{ik} v_i = (Mv)_k + (M^T v)_k \stackrel{M=M^T}{=} 2(Mv)_k.$$

In vector notation: $\nabla_v (v^T M v) = 2Mv$.

Appendix C: The Graph Laplacian is Positive Semi-Definite

Definition (Positive Semi-Definite, PSD). A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is *positive semi-definite* (written $A \succeq 0$) if

$$f^T A f \geq 0 \quad \text{for all } f \in \mathbb{R}^n.$$

Equivalently, all eigenvalues of A are non-negative. PSD matrices arise naturally as objective Hessians or, as here, as operators whose quadratic form counts a non-negative weighted sum.

Claim. $L = D - W \succeq 0$.

Proof. For any $f \in \mathbb{R}^n$, from Step 3 we already showed

$$f^T L f = \frac{1}{2} \sum_{i,j} w_{ij} (f_i - f_j)^2.$$

Since $w_{ij} \geq 0$ and $(f_i - f_j)^2 \geq 0$ for all i, j , every term is non-negative, so $f^T L f \geq 0$. Hence $L \succeq 0$.

Consequence. All eigenvalues of the generalised problem $Lf = \lambda Df$ are non-negative (given $D \succ 0$), and the smallest is $\lambda_0 = 0$ with eigenvector $\mathbf{1}$.